

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION**Revision Test 2026****Chemistry — Class 9 (Science Group)**

Syllabus: Chapter 1 - Nature of Chemistry in Science | Chapter 2 - Matter | Chapter 3 - Atomic Structure | Chapter 4 - Periodic Table & Periodicity of Properties | Chapter 5 - Chemical Bonding | Chapter 6 - Stoichiometry | Chapter 7 - Electrochemistry | Chapter 8 - Energetics | Chapter 9 - Chemical Equilibrium

Total Marks: 71**Time: 2 Hours 15 Minutes****Paper: Exam-20 (Version 1)**

1. Attempt **ALL** questions. There are no choices in this paper.
2. Section A (60 marks): 60 MCQs. Fill the correct bubble. Suggested time: 60 minutes.
3. Show complete working for every calculation. Always write the correct unit with your final answer.
4. Use black or blue ink only. Pencil is allowed only for diagrams. Relative atomic masses are given where needed.

SECTION A — Multiple Choice Questions (60 × 1 = 60 Marks)

#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	Which branch of chemistry investigates the essential question: "How do biomolecules such as carbohydrates, proteins, nucleic acids and lipids contribute to the structure and function of living organisms?"	analytical chemistry	polymer chemistry	biochemistry	geochemistry	(A)(B)(C)(D)
2	"How do geological processes influence the distribution of elements in the Earth's crust?" is an essential question of:	astrochemistry	environmental chemistry	geochemistry	industrial chemistry	(A)(B)(C)(D)
3	"How do nuclear reactions differ from chemical reactions, and what are their applications?" is an essential question of:	physical chemistry	organic chemistry	medicinal chemistry	nuclear chemistry	(A)(B)(C)(D)
4	The branch of chemistry that deals with the chemical processes and reactions occurring in astronomical environments such as stars, planets and comets is:	geochemistry	astrochemistry	environmental chemistry	industrial chemistry	(A)(B)(C)(D)
5	A solution of any metal in liquid mercury is called:	an alloy	an amalgam	a colloid	a suspension	(A)(B)(C)(D)
6	Fog, clouds and mist are solutions in which the solute is liquid water and the solvent is:	a gas	a solid	a liquid	a plasma	(A)(B)(C)(D)
7	In the ghee industry hydrogen gas is dissolved on the surface of solid nickel. This mixture is an example of:	a solid solution	a gaseous solution	a liquid solution	a suspension	(A)(B)(C)(D)
8	The solubility of a solute depends on the solvent used, the temperature and the:	colour of the solute	size of the container	pressure	shape of the container	(A)(B)(C)(D)
9	A carbonated drink, in which carbon dioxide is dissolved in water, is an example of:	a liquid dissolved in a gas	a gas dissolved in a liquid	a solid dissolved in a liquid	a solid dissolved in a solid	(A)(B)(C)(D)
10	An atomic model is best described as:	an exact photograph of a single atom	a simplified representation that helps us to understand the structure of an atom	a machine used for splitting atoms	an instrument for measuring atomic mass	(A)(B)(C)(D)
11	Tritium is not found in naturally occurring hydrogen because its nucleus is:	highly unstable	too small	electrically neutral	extremely heavy	(A)(B)(C)(D)
12	Radioisotopes are isotopes that	an equal	no neutrons at	more	an unstable	(A)(B)(C)(D)

	have:	number of protons and neutrons	all	electrons than protons	combination of neutrons and protons	
13	The number of neutrons in one atom of the isotope ^{80}Br (atomic number 35) is:	35	80	115	45	(A)(B)(C)(D)
14	When a nucleus emits a proton during radioactive decay, the atom is changed into:	another isotope of the same element	another element	an ion of the same element	a molecule of the same element	(A)(B)(C)(D)
15	The strong nuclear force that binds the nucleus together acts between:	protons and electrons only	neutrons and electrons only	electrons and electrons only	proton and proton, neutron and neutron, and neutron and proton	(A)(B)(C)(D)
16	Protons and electrons moving at the same speed are passed through a uniform electric field. The electrons are deflected far more strongly than the protons because the electron's:	mass is very much smaller	charge is very much larger	speed increases in the field	charge is positive	(A)(B)(C)(D)
17	In Rutherford's planetary model of the atom, the electrostatic force of attraction between the nucleus and a revolving electron is balanced by the:	centrifugal force of the revolving electron	strong nuclear force	gravitational force of the nucleus	magnetic force of the nucleus	(A)(B)(C)(D)
18	According to the Auf Bau principle, the sub-shell filled immediately after the 3p sub-shell is:	3d	4s	4p	4d	(A)(B)(C)(D)
19	An atom forms an anion by gaining one or more electrons. This happens most readily when the atom has a relatively high:	ionization energy	atomic radius	shielding effect	electron affinity	(A)(B)(C)(D)
20	The block to which an element belongs is decided by:	the number of neutrons in its nucleus	the sub-shell in which its last electron is placed	its relative atomic mass	the total number of its shells	(A)(B)(C)(D)
21	An element whose electronic configuration is $1s^2 2s^2 2p^2$ belongs to:	Group 2	Group 4	Group 14	Group 12	(A)(B)(C)(D)
22	The elements of Group 16 of the periodic table are collectively known as the:	halogens	alkali metals	chalcogens	noble gases	(A)(B)(C)(D)
23	Across a period of the periodic table, the density of the elements:	increases to a maximum near the middle of the period and then decreases	decreases steadily from left to right	remains exactly the same throughout	increases steadily right up to Group 18	(A)(B)(C)(D)
24	The first ionization energies of four elements are Na = 496, Mg = 738, Al = 578 and Si = 786 kJ/mol. The element that holds its outermost electron most tightly is:	Na	Mg	Al	Si	(A)(B)(C)(D)
25	The hydrogen halide with the greatest thermal stability is:	HCl	HBr	HI	HF	(A)(B)(C)(D)
26	Transition elements have high melting points mainly because of:	their unusually large atomic radii	their very low densities	stronger metallic bonding arising from partially filled d sub-shells	the complete absence of free electrons	(A)(B)(C)(D)
27	The noble gases are present in the atmosphere as:	diatomic molecules	triatomic molecules	giant lattices	mono-atomic gases	(A)(B)(C)(D)
28	All of the halogens exist as:	mono-atomic gases	ionic solids	diatomic molecules	metallic lattices	(A)(B)(C)(D)
29	Reactivity increases on going down	sodium	lithium	potassium	rubidium	(A)(B)(C)(D)

	Group 1, so the least reactive alkali metal is:					
30	The period number of an element is the same as:	the number of its valence electrons	the number of protons in its nucleus	the principal quantum number (n) of its valence shell	its group number	(A)(B)(C)(D)
31	Elements whose atoms have five, six or seven valence electrons usually react by:	losing electrons to form cations	gaining electrons to form anions	sharing electrons with noble gases	losing protons from the nucleus	(A)(B)(C)(D)
32	Non-metals generally have lower melting and boiling points than metals because their atoms and molecules are held together by:	weaker forces such as covalent, van der Waals or hydrogen bonds	strong metallic bonds	strong ionic bonds	strong nuclear forces	(A)(B)(C)(D)
33	Carbon dioxide contains strong covalent bonds inside each molecule, yet it has a very low melting and boiling point. This is because:	its covalent bonds break very easily	it contains free electrons	the forces between its separate molecules are weak	it forms a giant covalent structure	(A)(B)(C)(D)
34	In a molecule of hydrogen cyanide, HCN, the bond joining the carbon atom to the nitrogen atom is a:	single covalent bond	triple covalent bond	double covalent bond	coordinate covalent bond	(A)(B)(C)(D)
35	Boron trifluoride, BF ₃ , is described as electron deficient because the boron atom has only:	one pair of electrons in its valence shell	three pairs of electrons in its valence shell	two pairs of electrons in its valence shell	four pairs of electrons in its valence shell	(A)(B)(C)(D)
36	According to the textbook, the adhesive action of paints and dyes is developed because of:	ionic bonding	metallic bonding	hydrogen bonding	coordinate covalent bonding	(A)(B)(C)(D)
37	The two-dimensional layers of graphite are held to one another by:	weak van der Waals forces	strong covalent bonds	ionic bonds	metallic bonds	(A)(B)(C)(D)
38	A beryllium atom, 1s ² 2s ² , loses two electrons. The ion formed has the electronic configuration of:	hydrogen	neon	helium	argon	(A)(B)(C)(D)
39	The textbook gives three factors that determine the properties of a substance: the type of particles present, the way the particles are connected to each other, and:	the colour of the substance	the price of the substance	the temperature of the room	the arrangement of the particles	(A)(B)(C)(D)
40	Metals can be hammered into thin sheets because, when a force is applied:	all the metallic bonds break at once	the free electrons are destroyed	the layers of metal cations slide over one another	the metal changes into an ionic compound	(A)(B)(C)(D)
41	Ionic compounds are strong in compression but they are also:	malleable	ductile	brittle	elastic	(A)(B)(C)(D)
42	Batteries and fuel cells make use of ionic compounds as:	catalysts	insulators	lubricants	electrolytes	(A)(B)(C)(D)
43	For which one of the following compounds are the empirical formula and the molecular formula exactly the same?	CH ₄	C ₆ H ₁₂ O ₆	H ₂ O ₂	C ₂ H ₆	(A)(B)(C)(D)
44	The formula mass of calcium hydroxide, Ca(OH) ₂ , is: (Ca = 40, O = 16, H = 1)	57 amu	74 amu	58 amu	96 amu	(A)(B)(C)(D)
45	The number of moles present in 1.5 kg of magnesium oxide, MgO, is: (Mg = 24, O = 16)	0.375 mol	3.75 mol	375 mol	37.5 mol	(A)(B)(C)(D)
46	The number of atoms present in 23 g of sodium is: (Na = 23)	2.3×10^{23}	6.022×10^{23}	3.011×10^{23}	1.204×10^{24}	(A)(B)(C)(D)
47	When the equation $__\text{Ca} + __\text{H}_2\text{O} \rightarrow __\text{Ca(OH)}_2 + __\text{H}_2$ is correctly balanced, the coefficients are:	1, 1, 1, 1	2, 2, 1, 2	1, 2, 1, 1	2, 1, 2, 1	(A)(B)(C)(D)

48	Ions that appear unchanged on both sides of an ionic equation and take no part in the reaction are called:	free radicals	complex ions	formula units	spectator ions	(A)(B)(C)(D)
49	The molecular formula of the compound whose structural formula is $\text{CH}_3\text{-CH}_2\text{-NH}_2$ is:	$\text{C}_2\text{H}_7\text{N}$	$\text{C}_2\text{H}_5\text{N}$	$\text{C}_2\text{H}_6\text{N}$	$\text{C}_3\text{H}_7\text{N}$	(A)(B)(C)(D)
50	The formula of the compound formed between calcium ions, Ca^{2+} , and phosphate ions, PO_4^{3-} , is:	CaPO_4	$\text{Ca}_2(\text{PO}_4)_3$	$\text{Ca}_3(\text{PO}_4)_2$	$\text{Ca}(\text{PO}_4)_2$	(A)(B)(C)(D)
51	The binary ionic compound formed between aluminium and nitrogen is named:	aluminium nitrate	aluminium nitrite	nitrogen aluminide	aluminium nitride	(A)(B)(C)(D)
52	The mass of 1.5 moles of nitrogen gas, N_2 , is: (N = 14)	42 g	14 g	21 g	28 g	(A)(B)(C)(D)
53	The number of molecules present in 0.5 moles of hydrogen peroxide, H_2O_2 , is:	3.011×10^{23}	6.022×10^{23}	1.204×10^{23}	3.011×10^{22}	(A)(B)(C)(D)
54	The oxidation number of nitrogen in nitric oxide, NO, is:	+1	+2	+3	+4	(A)(B)(C)(D)
55	In the reaction $\text{CO}_2 + 2\text{Mg} \rightarrow 2\text{MgO} + \text{C}$, the reducing agent is:	Mg	CO_2	MgO	C	(A)(B)(C)(D)
56	In a polyatomic (complex) ion, the sum of the oxidation numbers of all the atoms present is equal to:	zero	the number of atoms in the ion	the group number of the central atom	the charge on the ion	(A)(B)(C)(D)
57	In a redox reaction, the oxidizing agent or the reducing agent is:	the whole molecule or formula unit that contains the atom whose oxidation number changed	only the single atom whose oxidation number changed	always one of the products	always a catalyst	(A)(B)(C)(D)
58	In the change $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$, the silver ion is:	oxidized, because it gains an electron	reduced, because it gains an electron	oxidized, because it loses an electron	reduced, because it loses an electron	(A)(B)(C)(D)
59	A collision between reacting particles can be effective only if the energy of the colliding particles is high enough to overcome the:	force of gravity acting between them	attraction of the walls of the container	pressure exerted by the surroundings	repulsion between the electrons around the reacting particles	(A)(B)(C)(D)
60	A catalyst speeds up a reaction by providing a new mechanism of lower activation energy, but it does not change the:	overall enthalpy of the reaction	rate of the reaction	mechanism of the reaction	activation energy of the reaction	(A)(B)(C)(D)

SECTION B – Short Questions (2 × 3 = 6 Marks)

Q.2	Attempt both parts.	Marks
(i)	A compound of nitrogen and oxygen contains 30.4% nitrogen and 69.6% oxygen by mass. Its relative molecular mass is 92 . (N = 14, O = 16) (a) Determine the empirical formula of the compound, showing your working. [2] (b) Determine its molecular formula . [1]	3
(ii)	Ammonia, NH_3 , is a covalent compound. (Atomic numbers: N = 7, H = 1) (a) Draw a dot-and-cross diagram to show the bonding in one molecule of ammonia. [2] (b) State the number of bond pairs and the number of lone pairs of electrons around the nitrogen atom. [1]	3

SECTION C – Long Question (1 × 5 = 5 Marks)

Q.3	Attempt all parts.	Marks

Potassium sulphate, K_2SO_4 , is used as a fertilizer. (K = 39, S = 32, O = 16, H = 1)

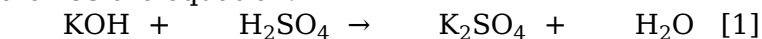
(a) Calculate the **formula mass** of potassium sulphate. [1]

(b) Calculate the **number of moles** present in 60.9 g of potassium sulphate. [1]

(c) Calculate the **number of formula units** present in that sample. [1]

(d) Potassium sulphate is prepared by reacting potassium hydroxide with sulphuric acid.

Balance the equation:



(e) Rewrite your balanced equation from part (d) with the correct **state symbols**, given that potassium hydroxide and sulphuric acid are used as aqueous solutions, potassium sulphate is formed in aqueous solution and water is a liquid. [1]

Chapters covered: Chemistry Class 9 — Chapter 1 (Nature of Chemistry in Science), Chapter 2 (Matter), Chapter 3 (Atomic Structure), Chapter 4 (Periodic Table & Periodicity of Properties), Chapter 5 (Chemical Bonding), Chapter 6 (Stoichiometry), Chapter 7 (Electrochemistry), Chapter 8 (Energetics), Chapter 9 (Chemical Equilibrium).

Candidate's Signature: _____

Invigilator's Signature: _____

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